

Xinyue (Margaret) Zhang

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Available for full-time internship (CPT): June 2026 – Mar 2027

Education

Northwestern University

Evanston, IL

M.S. in Computer Science — GPA: 3.67/4.0

Sep 2025 – Expected Jun 2027

– Coursework: Deep Learning for NLP, Scalable Software Architecture, Formal Verification, Multi-Agent Modeling Languages

University of Wisconsin – Madison

Madison, WI

B.S. in Computer Science — GPA: 3.66/4.0 · Dean's List 2021–2024

Sep 2021 – Dec 2024

– Coursework: Distributed Systems, Algorithms, Machine Learning, Database Applications

Professional Experience

Guoke Hengtai (Beijing) Medical Technology Co., Ltd.

Beijing, China

Software Developer Intern

May 2025 – Aug 2025

- Profiled and optimized hot-path warehouse-inventory queries on a medical-device platform with MyBatis composite indexes, cutting average query latency **30%** and improving operational efficiency for warehouse staff.
- Integrated a Redis caching layer via Spring Boot AOP @Cacheable to serve hot-path reads without hitting the database, reducing P99 response time **20%** at peak load.
- Diagnosed stale-read and oversell bugs from concurrent in-transit inventory updates; enforced **two-phase locking** at the database layer, validated with a multithreaded JUnit harness simulating **200 concurrent threads**, cutting worst-case latency **60%**.

Chanmao Song Network Technology Co., Ltd.

Urumqi, China

Software Developer Intern

May 2024 – Aug 2024

- Drove data-driven discovery — analyzed session logs and event-tracking data to surface **3 high-drop-off flows** — and led a **5-person team** to turn the findings into a prioritized engineering roadmap.
- Delivered the redesigned flows full-stack — rebuilt the Vue.js component layer and re-architected the REST API contracts — cutting user interaction steps **30%** and improving 7-day retention **40%**.
- Built a modular API integration layer and reusable JS test suite covering **12 critical user paths**, cutting post-launch regressions across Agile sprints.

Projects

Automated TA | Node.js, React/TypeScript, PostgreSQL, pgvector, Docker, LLM Integration

Sep 2025 – Jun 2026

- Shipped to production Fall 2025, serving **200+ Northwestern Medill students 24/7**; introduced an anonymous student→professor Q&A channel praised by the instructor and **extended to alumni**.
- Built a RAG pipeline: chunked material into 1,000-char windows, embedded into pgvector (cosine similarity), retrieved top-3 chunks per query, and assembled with 10-message history into structured LLM prompts to reduce hallucination.
- Architected multi-tenant infrastructure: user-scoped pgvector queries for data isolation, JWT-authenticated REST APIs with rate limiting, and a Dockerized full-stack deployment for CI/CD readiness.

KYMION | LangGraph, TLA+, TLC Model Checker, FastAPI, Redis

Apr 2026 – Present

- Built an **8-agent, 5-stage LLM orchestration engine** in LangGraph behind a FastAPI service, structuring each stage as a Work→Check→Resolve loop with a SQLite audit trail for end-to-end run observability.
- Engineered an LLM **reliability guardrail** that scores every agent output and vetoes any below a **0.8 confidence threshold**, bounding retries (3 redos / 5 disputes per stage) to keep unverified decisions from advancing the pipeline.
- Hardened correctness by specifying the workflow as a **TLA+ state machine** (5 safety invariants + liveness) mirrored state-for-state in the executable engine, validated by a **22-test suite** across 7 modules.

Biohybrid Robot: Using Plants as Sensors | Unity, Python, CO₂ Sensing, Embedded Systems

Jan 2026 – Present

- Replaced power-hungry electronic sensors with living plants as passive CO₂ light detectors — enabling **self-sustaining, zero-power sensing** that could make robots viable in remote or resource-scarce environments.
- Characterized the plant CO₂ signal and implemented infotaxis-style navigation in Unity; actively **co-authoring a research paper** on bio-integrated robotics and low-power sensing at scale.

Technical Skills

Languages: Python, Java, C/C++, C#, JavaScript/TypeScript, SQL, Go, HTML/CSS, NetLogo

Infrastructure: AWS, Docker, Kafka, Redis, Hadoop/HDFS, gRPC/Protobuf, Spark, GitHub Actions, PostgreSQL, MySQL, Git

Frameworks: React, React Native, Expo, Vue.js, Node.js, Spring Boot, pgvector, Parquet/PyArrow, TLA+, Lean 4, Agile/Scrum

AI / ML: PyTorch, TensorFlow, OpenAI API, LangChain, RAG, LLM Integration, Hugging Face, Scikit-learn